

Exploring Predictors of Bullying Perpetration Among Adolescents Using Machine Learning Approach

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Abstract

This study used machine learning methods to detect risk and protective factors for bullying perpetration in adolescents. The study sample consisted of 777 students with an age range of 11 to 16 years old. Multidimensional data covering both individual and environmental levels were collected. Individual factors included moral disengagement, normative beliefs about aggression, neuroticism, and self-control; environmental factors included parent–child relationships, deviant peer affiliation, school connection, and violent media exposure. The current study tested and compared six machine learning algorithms: Logistic Regression, Random Forest, Gradient Boosting Decision Tree, XGBoost, LightGBM, and Stacking, to detect risk and protective factors for bullying behavior. The results demonstrated that: (a) the Random Forest algorithm performed optimally, with recall, F1 score, and area under the curve values of 0.9394, 0.8516, and 0.8043, respectively; (b) both Gini

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importance and SHapley Additive exPlanations (SHAP) values identified self-control as the most significant protective factor, while moral disengagement was identified as the most influential risk factor. The recommended model not only provides an application value in preventing bullying but also provides a scientific basis for developing targeted interventions.

Keywords

machine learning, SHAP, adolescent bullying perpetration, risk and protective factors, self-control, moral disengagement

Introduction

Traditional bullying is intentional and repeated aggressive behaviors perpetrated by a peer or a group of peers that involve a power imbalance (Olweus, 1993) and can be verbal, relational, or physical forms. With the widespread use of the Internet, a new form of cyberbullying has also emerged (Slonje et al., 2013). Previous research has observed a significant overlap between traditional bullying and cyberbullying (Kowalski & Limber, 2013). Both forms share three core characteristics: intentionality, repetition, and a power imbalance between the perpetrator and the victim (Gini et al., 2018; Kowalski et al., 2012). Therefore, this study defines bullying as encompassing cyberbullying, conceptualizing it as intentional and repeated acts of aggression that can manifest in physical, verbal, relational, and electronic forms (Waasdorp & Bradshaw, 2015). A global school-based student health survey based on 191,228 students aged 12 to 15 years (48.9% boys) in 29 countries found that the mean prevalence of bullying victimization was 39.4% (Smith et al., 2023). Researchers found that bully-victim involvement is linked to higher levels of anxiety, depression symptoms, non-suicidal and self-injury (Balluerka et al., 2023; Liang et al., 2023; Ye et al., 2023), even increases difficulties for adult quality of life (Bettencourt et al., 2023).

Advantages of Machine Learning

Several previous empirical studies have explored the factors influencing bullying behavior. Still, these studies have primarily utilized classical statistical analyses (e.g., descriptive statistics and logistic regression analyses) to explore linear relationships between predictors and outcomes (Acosta et al., 2020). The most relevant limitation of classical statistical analysis is that simple regression fitting methods cannot assume nonlinear relationships

(Sajeev et al., 2021), often oversimplify the association between risk/protective factors and outcomes, and may reduce predictive accuracy. Additionally, simple regression methods can struggle with multicollinearity, a common issue in behavioral research, where predictors are highly correlated. This can lead to inflated standard errors and unreliable estimates of predictor importance. In contrast, machine learning algorithms, such as Random Forest, Gradient Boosting Decision Tree (GBDT), and XGBoost, can achieve the same tasks as simple regression but have unique advantages in addressing multicollinearity. These algorithms can effectively manage correlated predictors, providing more accurate and stable rankings of predictor importance, even when the features are highly interrelated, as is the case in this study. Machine learning is an emerging data analysis method in the age of digital intelligence and a core method of artificial intelligence, which relies on computer algorithms and aims to simulate human behavior through data analysis and modeling. Machine learning has many advantages over traditional statistical modeling. The processing of raw data by machine learning helps to improve the external validity of the final model. Traditional analysis methods (e.g., generalized linear models, generalized estimating equations, etc.) usually include all the data into the analysis uniformly, which often leads to overfitting of the model, and the generalization ability of the obtained model, that is, the external validity is poor. In contrast, machine learning divides the data into two subsets: the training set and the test set. The training set is used to fit the model, and the test set is used to test the model. The two subsets are then reorganized and re-divided randomly into a new training set and a new test set for model fitting and validation. This process is repeated several times and produces models with multiple predictable outcome variables. This validation enables an assessment of the external validity of the model (Campbell, 1986). Also, machine learning can quantify the feature importance of different influences in the overall model through several different. Such as Gini importance and the Shapley value (Shapley, 1953; Smith & Alvarez, 2021). Finally, machine learning can quantify the predictive effects of all predictor variables on an outcome variable to create a predictive model of the outcome variable (e.g., “the probability that a youth will bully others”). Once the model is created, the likelihood that a youth will bully others (i.e., the probability of bullying) can be output if the youth’s scores on the appropriate predictor variables are entered into the model. Based on the results, it is possible to make a preliminary prediction of the likelihood that a student will bully others in the future so that teachers or parents can pay attention to the problem and take targeted educational and intervention measures in light of the student’s actual performance. In recent years, many psychological researchers have applied machine learning to research areas such as cyber/

traditional bullying victimization (Yan et al., 2023), mental health strategy selection (Lamb et al., 2024), and suicidal ideation (Heckler et al., 2021). Yan et al. (2023) utilized six machine learning algorithms (Logistic Regression, Naive Bayes, Decision Tree, Random Forest, K-Nearest Neighbors (KNN), and Light Gradient Boosting Machine) to evaluate 40 potentially relevant factors in predicting traditional bullying victimization and cyberbullying victimization among Chinese adolescents. Zhu et al. (2025) used the light gradient boosting machine (LightGBM) to identify 30 potential factors in cyber-aggression. In contrast to these studies, which relied on a large number of features, the proposed study aims to use a significantly reduced number of features in training the machine learning model. By reducing the number of required features, the trained model will be able to predict outcomes with minimal data collection, making it more efficient and easier to implement in real-world settings.

Predictors of Bullying Behavior

Based on the advantages of machine learning mentioned above, this study will focus on identifying and ranking the importance of various risk and protective factors in bullying. The General Aggression Model (GAM), which asserts that aggression, including bullying behavior, results from the interaction of individual and environmental factors (Allen et al., 2018). Inspired by this framework, this study selected variables from both individual and environmental factors that are significantly associated with bullying behavior. Based on previous literature (Evans et al., 2013; Xiong et al., 2024), these factors were chosen according to the following principles: (a) Relevance: The selected predictors have been shown in prior research to be significantly associated with bullying behavior. (b) Representativeness: These predictors are widely used in previous studies on bullying and represent corresponding contexts; (c) Uniqueness: The selected factors are distinct and do not overlap significantly with one another. Specifically, for environmental factors, we chose to involve family, peer, school, and violent media exposure. For individual factors, we chose moral disengagement, normative beliefs about aggression, neuroticism, and self-control.

Family Factors

Family, particularly parent–child relationships, are foundational in shaping children’s social behaviors and emotional responses (Laursen & Collins, 2009). Family conflict and poor relationships with parents are risk factors for the development of delinquent behavior and bullying in children (Nocentini et al., 2019).

Maintaining warm and positive long-term relationships with parents may be the most critical family factor in developing harmonious peer relationships in young children and adolescents (Miller-Graff et al., 2016; Pavlova et al., 2016).

Peers Factors

The influence of peers on bullying is tremendous, especially in adolescence (Salmivalli et al., 2021). According to social learning theory, adolescents may engage in similar behaviors by observing and imitating the behaviors of their peers (Bandura & Walters, 1977). Empirical studies have found that deviant peer affiliation is a significant risk factor for bullying among adolescents (Gao et al., 2022; Wang et al., 2020, 2022).

School Factors

School is another important factor contributing to bullying. Research has shown that poor teacher–student are risk factors for aggression (Krause & David Smith, 2023). Also, when students have a low level of school connection (i.e., the degree to which they feel cared for and belong to the school), the higher the likelihood of involved bullying (Lucas-Molina et al., 2022).

Violent Media Exposure

Also, violent media exposure is an important environmental variable. Extensive longitudinal and experimental studies have consistently indicated that prolonged engagement with violent media correlates with an increase in physical aggression among adolescents (Delhove & Greitemeyer, 2021; Zhang et al., 2021). A large-scale meta-analysis of 136 scholarly articles indicates that exposure to violent video games can increase aggression, including aggressive behaviors, cognition, and emotions (Anderson et al., 2010). Such findings support the GAM, which posits a causal relationship between exposure to media violence and subsequent aggressive behaviors (Anderson & Bushman, 2002). However, this perspective was challenged by Ferguson et al. (2008) who proposed the Catalyst Model. According to the Catalyst Model, media violence acts more as a stylistic catalyst rather than a direct cause. The model suggests that individuals are not passively influenced by media; rather, they selectively imitate media violence based on their pre-existing tendencies. Hence, while media violence can serve as a stylistic enhancer, the primary determinants of aggressive behavior are rooted in factors like family violence and personal traits (Ferguson et al., 2008). It is essential to clarify that the focus of this study is not to explore the direct

causality between violent media and bullying behaviors but rather to ascertain if violent media constitutes an important feature of bullying. Therefore, violent media remains a critical environment variable in our analysis.

Normative Beliefs About Aggression

Normative beliefs about aggression refer to an individual's perception of aggressive behaviors as acceptable and justified means of resolving conflicts, believed to be endorsed by the majority (Hilvert-Bruce, & Neill, 2020). These beliefs form a crucial component of an individual's cognitive structure and play a significant role in influencing behavior, particularly in contexts of conflict resolution. According to Social Information Processing theory (Dodge & Crick, 1990), individuals with high normative beliefs about aggression are prone to biases in recognizing and interpreting social cues. They often encode situational cues as hostile, leading to misinterpretations of environmental signals. This misprocessing can escalate to aggressive actions as these individuals are more likely to respond based on these skewed perceptions (Smeijers et al., 2020). Empirical research supports this relationship, demonstrating that normative beliefs about aggression significantly predict aggressive and bullying behaviors (Burton et al., 2013; Zhu et al., 2020).

Moral Disengagement

Moral disengagement refers to a cognitive tendency wherein individuals rationalize unethical behaviors by absolving themselves of moral responsibility. This process involves redefining immoral actions to minimize their perceived harmfulness, vigorously denying personal accountability for the consequences of these actions, and actively suppressing empathy toward victims. Bandura et al. (1996) conceptualized moral disengagement as a cognitive process, suggesting that it impairs an individual's awareness of moral responsibility, motivation for moral conduct, and capacity for moral judgment. Also, it diminishes the experience of negative moral emotions during unethical behaviors, such as feelings of self-reproach or guilt. Some longitudinal studies have found moral disengagement is a significant predictor of more frequent bullying behaviors (Thornberg, 2023). A meta-analysis found that there was a significant positive correlation between moral disengagement and aggressive behaviors (Gini et al., 2014).

Neuroticism

Neuroticism, a sub-dimension of the Big Five personality model, reflects an individual's tendency to experience negative emotions (Barlow et al., 2014;

Ormel et al., 2013). Previous theoretical and empirical studies have suggested that neuroticism may be an important personality variable that causes aggression and is a risk factor for aggressive behavior. For example, a short-term longitudinal research showed high neuroticism was associated with a larger increase in physical aggression and a smaller decrease in verbal aggression over time (Hu et al., 2022). A study involving 259 men, including 39 incarcerated violent offenders with a broad range of trait aggression, found that those high in neuroticism are at a higher risk of exhibiting aggressive behavior (Dam et al., 2021).

Self-Control

Self-control refers to an individual's ability to autonomously regulate behavior to conform to self or social expectations and to make long-term plans (Baumeister et al., 2007). It is also essential for adolescent development. Self-control theory suggests that self-control contains five domains: a general capacity for self-discipline, inclination toward deliberate/non-impulsive action, healthy habits, self-regulation in service of a work ethic, and reliability; these domains reflect the cognitive control and response inhibition functions of the human psyche (Berkman et al., 2011; Lopez et al., 2014). Longitudinal research and meta-analyses have all confirmed that individuals with low self-control are more likely to engage in aggression and bully others (Baumeister & Boden, 1998; Pung et al., 2015).

Current Study

This study will analyze the dataset using a variety of machine learning algorithms to predict adolescent bullying behavior. These algorithms include Logistic Regression, bagging integrated Random Forest, boosting integrated GBDT, XGBoost, LightGBM, and Stacking, where the base learners are Logistic Regression, GBDT, XGBoost, LightGBM, Support Vector Machine (SVM), and KNN, and the meta-learner is Logistic Regression to find the best-performing model to analyze our dataset. Logistic regression is a linear model suitable for simple relationships but struggles with capturing non-linear patterns, which are often present in models predicting bullying behavior. Random Forest and GBDT, as ensemble methods, excel at modeling high-dimensional data with interactions between multiple predictors, which are common in bullying behavior research. XGBoost and LightGBM, both efficient versions of GBDT, are particularly good at handling non-linearity and complex interactions. XGBoost is known for its regularization capabilities, which help to prevent overfitting and improve model generalization. LightGBM, on the other hand, is optimized for faster training and lower

memory usage, while maintaining high predictive performance. Stacking, by combining multiple base models, leverages the strengths of each individual model to improve overall predictive accuracy. Given the complex nature of bullying behavior, which involves both individual and environmental factors, combining different algorithms could enhance predictive performance, providing more reliable results for intervention and prevention strategies. Then, all the factors that may lead to bullying behavior are identified and ranked according to the importance of the features, respectively.

Method

Participants

Participants were students from two middle schools in China. This survey was approved by the Ethics Committee of the author's affiliated institution. The data were collected during September 2021. Before the surveys, informed consent was obtained from parents, students, and the schools. A total of 820 participants completed the survey. Data from participants with substantial missing values or those who exhibited response patterns indicating lack of engagement (e.g., consistent identical answers across items) were excluded. Finally, we are collected 777 valid questionnaires with 94.76% efficiency. The sample comprised 336 girls and 441 boys, 16.60% of whom were only children—the age of participants between 11 and 16 ($M = 13.57$, $SD = 0.98$).

Instruments

The survey used Chinese version of the Olweus Bully/Victim Questionnaire (Zhang et al., 1999) and Cyberbullying Questionnaire, developed by Li (2020) to assess the extent to which participants had recently experienced bullying. The Chinese version of the Olweus Bully Questionnaire contains six items (e.g., I once spread rumors about my classmates and tried to make others dislike him [her]). Items were scored using a 4-point scale that ranges from 1 = *never*; 2 = *once in a while*; 3 = *now and then*; 4 = *often*. Also, a categorical variable (indicating bullied or not bullied) was computed. Based on prior work (Olweus, 2012), a student is classified as bullied if she or he reported experiencing once in a while or more. In Cyberbullying Questionnaire, a representative sample item is "I spread others' privacy information on the Internet." Participants were asked to report how frequently they engaged in a range of online bullying behaviors using a 4-point Likert scale (1 = *It has never happened to me*; 2 = *It happened 1–2 times*; 3 = *It happened 3–5 times*; 4 = *It happened more than 5 times*). As above, a categorical

variable (indicating bullied or not bullied) was computed. A student is classified as bullied if she or he reported experiencing one to two times cyberbullying or more (Olweus, 2012). Therefore, bullying was recoded as two binary variables of adolescents with outcomes “have bullied others” (score of 1)/“have not bullied others” (score of 0).

The predictor variables include parent–child relationship, deviant peer affiliation, school connection, media violence exposure, moral disengagement, normative beliefs about aggression, neuroticism, and self-control. A summary of the measurement scales, scoring methods, and Cronbach’s α values for these variables is presented in Table 1. For a detailed summary of these scales, please refer to the Supplemental Material. Additionally, demographic information, including gender, was collected as a categorical variable to examine its potential factors on bullying behavior.

Statistical Analysis

The dataset utilized for machine learning analysis comprised of 777 participants. The dataset was split into training and testing sets in a 7:3 ratio, a common choice to avoid underfitting while ensuring a sufficiently large test set for model evaluation (e.g., Chen et al., 2024; Yan et al., 2023). Considering the diversity of machine learning algorithms and the unknown relationships between predictor variables and outcome variables, we employed a variety of machine learning algorithms to analyze the dataset and predict adolescent bullying behavior. These algorithms included Logistic Regression, bagging integrated Random Forest, boosting integrated GBDT, XGBoost, LightGBM, and Stacking, where the base learners are Logistic Regression, GBDT, XGBoost, LightGBM, SVM, and KNN and the meta-learner is Logistic Regression. To optimize the models, we used Grid Search and Cross-Validation methods to tune the parameters of each model and determine the best-performing model.

The statistical analysis and machine learning modeling were performed using Python 3.8. The scikit-learn library was employed for machine learning algorithms, while SHapley Additive exPlanations (SHAP) analysis was conducted using the corresponding SHAP library. Data visualization and plotting were generated using seaborn and matplotlib libraries.

In this study, the performance of the machine learning models was assessed using the following evaluation metrics: *Accuracy*, the proportion of correctly predicted samples out of the total samples. *Precision*, the proportion of true positive predictions among all positive predictions made by the model. *Recall*, the proportion of true positive samples correctly identified by the model out of all actual positive samples. *F1 score*, the harmonic mean of

Table 1. Summary of Measures.

Variables	Measurement Scale	Scoring	Cronbach's α
Deviant peer affiliation	Deviant Peer Affiliation Scale (Fergusson & Horwood, 1999; Kendler et al., 2007)	Likert, 1–5	.79
Parent–child relationship	Parent-Child Relationship Scale (Furman & Buhrmester, 1985)	Likert, 1–5	.82
School connection	School Connection Scale (Resnick et al., 1997)	Likert, 1–7	.86
Moral disengagement	Moral Disengagement Scale (Bandura et al., 1996; Wang & Yang, 2010)	Likert, 1–5	.86
Neuroticism	NEO Five-Factor Inventory (Costa & McCrae, 1992)	Likert, 1–5	.86
Media violence exposure	Exposure to Violent Media Questionnaire (Li et al., 2015; Zhang et al., 2021)	Average of frequency (Likert, 1–5) $\times$ Violence Score (Likert, 1–5)	.82
Normative beliefs about aggression	Normative Beliefs About Aggression Scale (Huesmann et al., 1992)	Likert, 1–4	.74
Self-control	Self-Control Scale (Tangney et al., 2004; Yi, 2013)	Likert, 1–5	.77

precision and recall, providing a single metric that balances both precision and recall. Area under the curve (AUC), the area under the receiver operating characteristic (ROC) curve, representing the model's overall ability to distinguish between classes. A higher AUC indicates better model performance. ROC curve, a graphical tool for evaluating the performance of a classification model by plotting the true positive rate against the false positive rate. *Confusion matrix*, a matrix that describes the performance of the classification model by displaying the counts of true positive, false positive, true negative, and false negative predictions.

Subsequently, the best-performing algorithm was selected to establish the bullying prediction model. The relative importance of each feature (referred

to as factors in social science terminology) was determined for each specific algorithm. Feature importance is a key concept in machine learning, with various calculation methods available. The feature importance was explained using average SHAP values and Gini importance in this study. Gini Importance is a measure of feature importance in decision tree and random forest models. When constructing a decision tree, the Gini coefficient is used to measure how much a feature improves the purity of a dataset partition. The SHAP method calculates global interpretation by computing and combining SHAP values across the entire dataset, ensuring fair distribution of machine learning algorithm predictions among different features or variables. More and more researchers have started to conduct scientific studies on the relative contribution of predictor variables in predictive models with the SHAP value (Yan et al., 2023).

Results

Machine Learning Model Performances

Table 2 presents the model performance of six machine learning algorithms for predicting bullying behaviors. The Random Forest model demonstrates the best recall and F1 score. Additionally, the Random Forest model also provides relatively high accuracy and precision scores. Figure 1 shows the ROC curves for the various machine learning models, offering quantitative guidance for the machine learning algorithm performances by using the AUC score, that is, area under the curve. For the Random Forest algorithm, the corresponding AUC is the highest at 0.8043, indicating that this model has an 80.43% probability of correctly predicting whether an adolescent will engage in bullying behavior. Figure 2 shows the confusion matrix for various machine learning models. True negatives are correctly predicted non-bullying instances; false positives are incorrectly predicted bullying instances; false negatives are incorrectly predicted non-bullying instances; true positives are correctly predicted bullying instances. The Random Forest model appears to have the best performance with the highest true positives (155) and the lowest false negatives (10), indicating its effectiveness in predicting bullying behaviors while maintaining a balance with false positives.

Feature Importance to Predict the Bullying

Figure 3 displays the feature importance ranking from the Random Forest for predicting bullying using Gini importance. The figure ranks the importance of various features in decreasing order from top to bottom. In general,

Table 2. Machine Learning Algorithm Performances for Bullying.

Model	Accuracy	Precision	Recall	F1 Score	AUC
Logistic regression	0.7735	0.8043	0.8970	0.8481	0.7859
XG boost	0.7564	0.7727	0.9273	0.8430	0.7751
Random forest	0.7692	0.7789	0.9394	0.8516	0.8043
GBDT	0.7265	0.7440	0.9333	0.8280	0.7448
LightGBM	0.7393	0.7680	0.9030	0.8301	0.7560
Stacking	0.7735	0.7979	0.9091	0.8499	0.7812

Note. GBDT=Gradient Boosting Decision Tree; AUC=area under the curve.

self-control and moral disengagement are the most relevant features of Bullying. Figure 4 displays the feature importance ranking from Random Forest using SHAP values. Also, self-control and moral disengagement were consistently identified as the most important Features predicting bullying behavior. Figure 5 presents the feature importance within the Random Forest algorithm in predicting bullying using SHAP values. Figure 5 also reveals the direction of effects at different levels of each factor. The dot color is redder when the feature value increases, and the dot color is bluer when the feature value decreases. A greater SHAP value is associated with a greater impact on the model output. The results in Figure 5 show that higher values of moral disengagement are associated with a higher risk of being bullied, while higher values of self-control are associated with a lower risk of being bullied.

Discussion

Performance of Machine Learning Algorithms

The current study investigated the best-performing machine learning model for identifying risk and protective factors for bullying among adolescents. After comparing six machine learning algorithms, it was determined that Random Forest outperformed the others in assessing the importance of each factor in predicting bullying in our dataset. One of the key advantages of Random Forest is its ability to handle multicollinearity between variables. In psychological research, many factors are highly correlated, and this can affect the performance of other models. Random Forest does not require independent predictors and can effectively manage correlated variables, making it a strong choice when dealing with complex datasets where multicollinearity exists. In particular, this algorithm has a recall rate of 93.94%, which means that 93.94% of bullies are correctly identified. This high recall is particularly

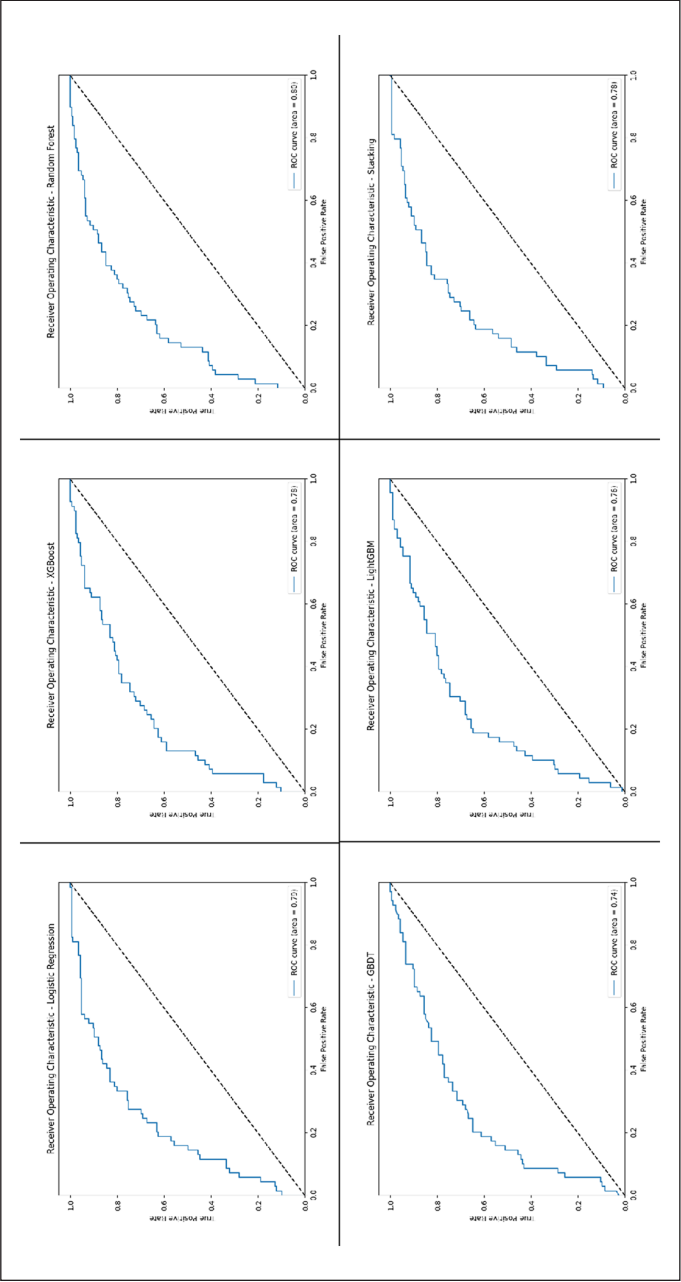


Figure 1. ROC curves for various machine learning models.
Note. ROC=receiver operating characteristic.

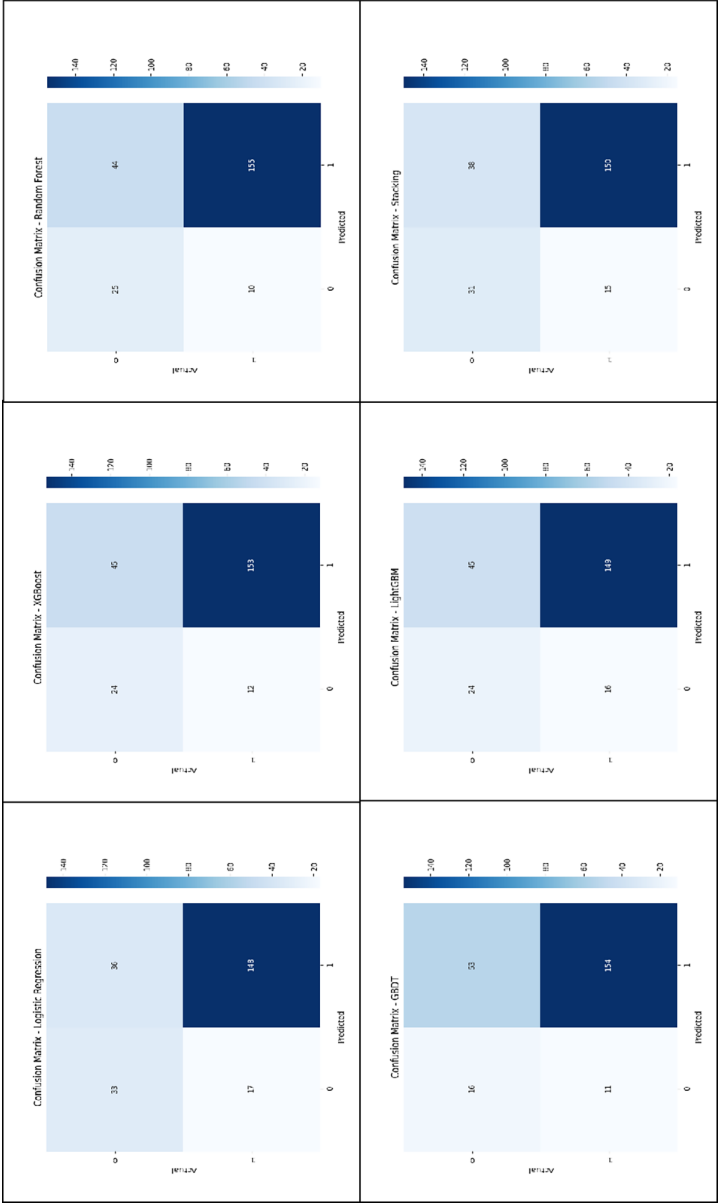


Figure 2. Confusion matrix for various machine learning models.

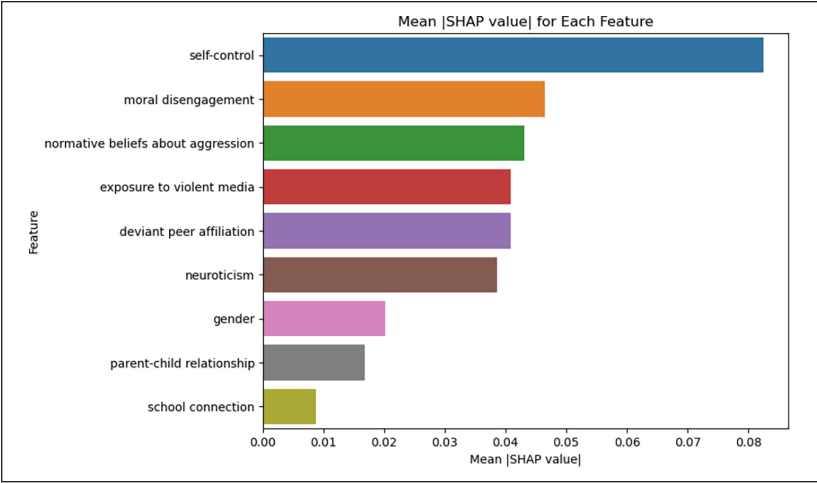


Figure 3. Feature importance ranking (Gini importance) from Random Forest.

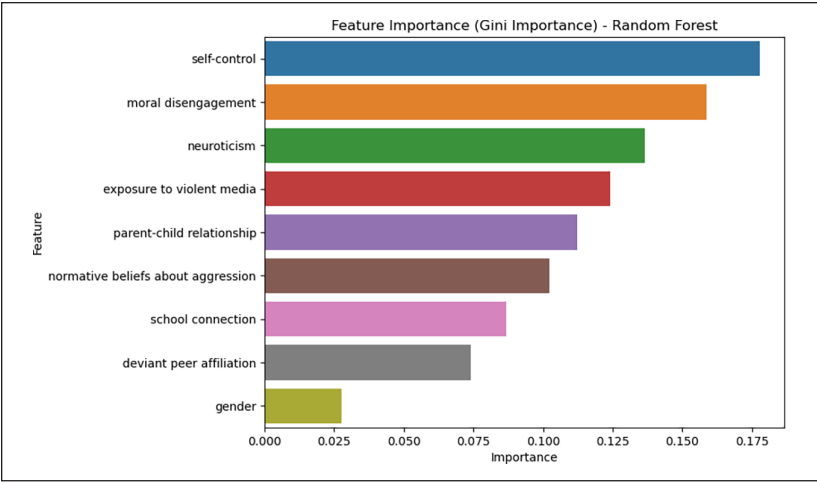


Figure 4. Feature importance ranking (SHAP values) from Random Forest.
Note. SHAP = SHapley Additive exPlanations.

important in the context of bullying prevention, as it ensures that a large proportion of potential perpetrators are flagged for further intervention. However, it is worth noting that we cannot “label” students based on the results of a

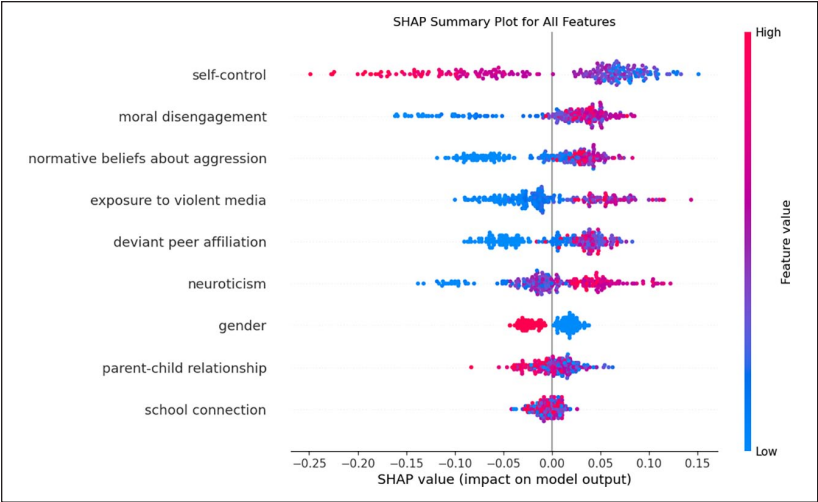


Figure 5. SHAP value for features in bullying from Random Forest.
Note. SHAP = SHapley Additive exPlanations.

single test, and that the use of predictive modeling results needs to be considered in the context of research ethics and schooling practices.

The Key Predictive Features

To interpret the feature importance within the Random Forest model, the study employed both traditional Gini importance and SHAP values. The combination of these methods provided a more comprehensive explanation of the results. Both methods indicated that self-control and moral disengagement were the most predictive factors for bullying behavior, with self-control being the most significant protective factor and moral disengagement the most influential risk factor. Specifically, among these factors, self-control was the most critical predictor of bullying behavior, significantly surpassing other protective factors. This finding is consistent with previous research (Boccio & Beaver, 2021; Fenny, 2022), which has indicated that low self-control is an important causal factor in aggressive behavior and crime. Also, this result is well-explained by Gottfredson and Hirschi's (1990) Theory of Low Self-Control, which posits that individuals with low self-control are more likely to engage in deviant behaviors, including bullying, due to their inability to delay gratification, regulate their impulses, or consider the long-term consequences of their actions. Numerous intervention studies support

the idea that enhancing self-control can reduce aggressive behaviors, including bullying (Kip et al., 2021; Zhang & Zhang, 2023). For example, Zhang and Zhang (2023) found that mindfulness-based interventions have been successful in decreasing aggression by improving self-control. These interventions underscore the potential of self-control as a protective factor in preventing bullying.

Similarly, moral disengagement was identified as the second most important factor for bullying behavior. These findings also align with Bandura's (2016) Social Cognitive Theory of Moral Disengagement, which suggests that individuals can engage in harmful behaviors, such as bullying, only if they can justify their actions as morally acceptable. Moral disengagement provides a cognitive distortion that allows individuals to rationalize their harmful behavior, adolescents can diminish the psychological barriers to aggression, making bullying behavior more likely. This finding aligns with the study by Zhang et al. (2023), which also used machine learning methods to rank the importance of various risk and protective factors in cyber-aggression. In their study, moral disengagement was ranked highly as well, underlining its central role in aggressive behaviors.

Implications

One of the implications of our research is the ability to use the nine identified features (indicators/factors) to recognize potential bullies among adolescents. Importantly, these features are relatively few, making the identification process both efficient and practical. With a relatively small dataset of 777 samples and only 9 features, we were able to develop a highly efficient model that provides a good balance of predictive accuracy and computational efficiency. By leveraging machine learning models trained on these features, schools and psychologists can develop predictive tools to flag high-risk individuals. This simplicity ensures that the model remains applicable in real-world settings, while still maintaining robust performance for bullying prevention. This proactive approach can significantly reduce the prevalence of bullying by addressing issues before they escalate. Our study found that self-control and moral disengagement are the most critical predictors of bullying behavior, it is essential to target this variable in practice. For example, experiments have demonstrated that school-based mindfulness intervention can reduce bullying by increasing teenagers' self-control (Liu et al., 2022). Also, the Bullying Literature Project-Moral Disengagement Version has demonstrated effectiveness in reducing moral disengagement and bullying (Wang & Goldberg, 2017). This study focuses on the use of three types of integrated learning methods, Bagging, Boosting, and Stacking, to improve the generalization

ability, robustness, and stability of the model. Our research demonstrated the superiority of models trained using ensemble learning techniques. Ensemble learning, which combines predictions from multiple base learners to form a robust and stable model, showed higher accuracy and reliability in predicting bullying behavior. Although ensemble methods generally require more training time, this is not a significant concern with smaller sample sizes typical in psychological studies. Thus, ensemble learning emerges as a highly suitable approach for analyzing psychological questionnaire data, given the typical data volumes and feature sets in this field. Psychological researchers can consider it as a go-to method for model training in the future.

Limitations

The current study sample consists of adolescents aged 11 to 16 from two middle schools in China, which provides insight into early adolescence within east Asian culture. However, the findings should be interpreted with caution when considering their applicability to other cultural or regional contexts, such as Western or African countries, where different educational systems, family structures, and social expectations may yield different patterns of behavior. Second, the reliance on self-reported data for measuring bullying behaviors could affect the false positives of the predictive model. In particular, the model predicts a student as a bully, but the student has not reported engaging in bullying behavior. Self-reports can be influenced by social desirability bias. Third, due to the “black box” nature of machine learning models, even though SHAP values are used to explain the contribution of each feature, they still do not provide a deeper understanding of the underlying factors driving bullying behavior. Therefore, more advanced interpretability methods could be explored in future research to gain a better insight into the complex mechanisms behind bullying perpetration. Last, this study conceptualized bullying as a broad construct encompassing both traditional bullying and cyberbullying, rather than treating them as separate behaviors. While this approach allows for a comprehensive understanding of bullying perpetration, it may overlook potential differences in risk/protective factors between the two forms. Future studies could further refine predictive models by examining traditional bullying and cyberbullying separately to identify unique predictors and intervention strategies.

Data Availability Statement

Data and the code of six machine learning algorithms are available on request from authors.

Declaration of Conflicting Interests

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Ethical Approval and Informed Consent Statements

All research procedures followed were in accordance with the ethical standards of the World Medical Association (Declaration of Helsinki) for experiments involving human subjects. All conducted procedures received approval from the ethical committee of the Academic Ethics and Moral Committee of Shanghai Normal University (Decision Number 097 of 2023).

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Supplemental Material

Supplemental material for this article is available online.

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